CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

Project Report | Python • PostgreSQL • Power BI

3,900	$59.76	3.75/5	$233,081
Customer records	Average purchase	Average rating	Total purchase revenue

1. Executive Summary

This project analyzes 3,900 customer shopping records to understand purchasing behaviour, category performance, subscription patterns, shipping preferences, discount usage, and age-group revenue. The workflow combines Pandas-based data preparation, PostgreSQL storage and SQL analysis, and a Power BI dashboard. The average purchase amount is $59.76, and the average review rating is 3.75 out of 5.

2. Business Objectives

• Measure overall customer activity and purchase value.

• Compare revenue across gender, product category, subscription status, shipping type, and age group.

• Explore discount usage, product ratings, repeat purchases, and popular products.

• Present the results in a clear interactive dashboard for business users.

3. Dataset Description

The source CSV contains 3,900 rows and 18 columns, including customer ID, age, gender, purchased item, category, purchase amount, location, size, color, season, review rating, subscription status, shipping type, discount/promo usage, previous purchases, payment method, and purchase frequency. Customer ages range from 18 to 70; the dataset contains 4 categories, 25 distinct items, and 50 locations.

4. Tools & Technologies

Tool	Purpose
Python / Pandas	Data loading, inspection, cleaning, feature engineering
Jupyter Notebook	Exploratory analysis and reproducible workflow
PostgreSQL / pgAdmin 4	Relational data storage and SQL querying
SQLAlchemy / psycopg2	Python connection to PostgreSQL
Power BI Desktop	Interactive dashboard and visual analytics

5. Data Preparation

• Inspected data types and missing values; 37 review-rating values were missing.

• Imputed missing review ratings using the median rating within each product category.

• Standardized column names to lowercase snake_case and renamed purchase amount to purchase_amount.

• Created age groups using quartile-based binning.

• Created a purchase-frequency-days feature by mapping frequency labels to approximate day intervals.

• Dropped promo_code_used after the notebook found it matched discount_applied for every row.

6. Key Findings

Revenue by gender

Male customer records contributed $157,890 in purchase revenue, while female customer records contributed $75,191. There are 2,652 male and 1,248 female records, so totals should be considered alongside record counts.

Revenue by category

Category	Records	Revenue	Avg. purchase
Clothing	1,737	$104,264	$60.03
Accessories	1,240	$74,200	$59.84
Footwear	599	$36,093	$60.26
Outerwear	324	$18,524	$57.17

Subscription comparison

Status	Records	Avg. purchase	Revenue
No	2,847	$59.87	$170,436
Yes	1,053	$59.49	$62,645

Subscribers averaged $59.49 per purchase compared with $59.87 for non-subscribers. This is a descriptive comparison only.

Shipping and age groups

Average purchase amount was $60.48 for Express shipping and $58.46 for Standard shipping. Age groups were generated from age quartiles; the Young adult group contributed $62,143, the highest among the four groups in this dataset.

7. Dashboard Overview

The Power BI dashboard includes customer count, average purchase amount, average review rating, subscription-status share, revenue and sales by category, and revenue and sales by age group. Slicers allow filtering by subscription status, gender, category, and shipping type.



Dashboard screenshot supplied with the project.

8. SQL Analysis

The SQL script is organized around ten business questions covering gender revenue, discount-driven spending, product ratings, shipping comparisons, subscription behaviour, discount rates, customer segments, top products per category, repeat buyers, and age-group revenue.

9. Limitations & Items to Verify

• The source CSV has 37 missing review ratings; the notebook imputes these using category medians.

• The notebook's frequency mapping contains label spelling/capitalization mismatches (for example, Annualy versus Annually), which may leave some engineered day values null.

• The supplied SQL script has an unfinished subscriber comparison query (Q5 uses the invalid token roavg and lacks the remaining query clauses).

• The discount-rate SQL expression should use numeric division/casting to avoid integer division.

• The analysis is descriptive and does not establish causal relationships.

10. Reproduction Steps

1. Install pandas, SQLAlchemy, psycopg2-binary, and Jupyter.

2. Create a PostgreSQL database named customer_behaviour.

3. Update the notebook's CSV path and local PostgreSQL connection settings.

4. Run the notebook to clean the data and load the customer table.

5. Run the SQL analysis script in pgAdmin 4 after correcting the incomplete query.

6. Connect Power BI to the PostgreSQL customer table and refresh the report.

11. Security & Publishing Checklist

The notebook provided for this report contains a hard-coded PostgreSQL password. Change that password before publishing, remove the credential from the notebook, clear any sensitive outputs, and use environment variables or a local configuration file excluded from Git. Do not commit database passwords.

12. Conclusion

This project demonstrates a complete beginner-to-intermediate analytics workflow: preparing raw customer data with Python, storing it in PostgreSQL, querying business questions with SQL, and communicating patterns through Power BI. The results provide a descriptive overview of customer and category behaviour that can support further business investigation.

Prepared from the supplied notebook, SQL script, CSV dataset, and Power BI dashboard screenshot.